

Total No. of Questions : 6]

SEAT No. :

P67

OCT. -16/BE/Insem. - 121

[Total No. of Pages : 2

B.E. (Mechanical)

GAS TURBINE AND PROPULSION

(2012 Course)(Elective-II)(Semester-I) [402045-A]

Time :1Hour]

[Max. Marks :30

Instructions to the Candidates:

- 1) Answer Q. 1 or Q. 2, Q. 3 or Q. 4, Q. 5 or Q. 6.
- 2) Draw suitable neat diagrams, wherever necessary.
- 3) Assume suitable data, if required.

Q1) a) State the principle of rocket propulsion with propellants and its properties. [5]

b) Explain the gas turbine design procedure. [5]

OR

Q2) a) Explain the aircraft propulsion with examples. [5]

b) Explain with neat sketch simple gas turbine power plant. [5]

Q3) a) Explain the polytropic efficiency of compressor and turbine with derivation. [4]

b) In a gas turbine, the pressure to which air at 15°C is compressed is 6bar. The same air is then heated to a maximum permissible temperature of 760°C. first in a heat exchanger and then combustion chamber. It is then expanded in two stages such that the expansion work is maximum. The air is reheated to 760°C after the first stage. Determine the cycle thermal efficiency, the work ratio (W_N/W_T) and net shaft work per kg of air. Intake pressure is 1 bar [6]

OR

Q4) a) Explain the intercooled compression Cycle with heat exchanger with derivation. [5]

b) A gas turbine operates on a pressure ratio of 6. the inlet air temperature to the compressor is 300 K and the air entering the turbine is at a temperature of 580°C. If the volume rate of air entering the compressor is 240 m³/s. Calculate the net power output of the cycle in MW. Also compute its efficiency. Assume that the cycle operates under ideal conditions. Compressor inlet press is 1bar [5]

P.T.O.

Q5) a) State the environment issues of Gas turbine. [4]

b) Determine the specific work output, specific fuel consumption and cycle efficiency for a simple cycle gas turbine with a free power turbine given the following specification: [6]

Compressor pressure ratio	12.0
Turbine inlet temperature	1350K
Isentropic efficiency of compressor,	0.86
Isentropic efficiency of each turbine	0.89
Mechanical efficiency of each shaft	0.99
Combustion efficiency	0.99
Combustion chamber pressure loss	6% of compressor delivery press.
Exhaust pressure loss	0.03bar
Ambient conditions, P_a , T_a	1 bar, 288K

OR

Q6) a) Explain with neat sketch thermodynamic analysis of turboprop engine.[5]

b) A turbojet engine draws air at the rate of 1 kg/s while flying at a speed of 900kmph. The velocity of gases at the exit of the nozzle is 620m/s. The engine uses fuel at the rate of 0.0125kg/s of calorific value 45100 kJ/kg. Find: [5]

(i) Fuel air ratio

(ii) Fuel consumption in kg/hr.

(iii) Thrust, thrust power and TSFC

(iv) Propulsive power and propulsive efficiency

(v) Thermal and overall efficiency of turbojet

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